HARD

PROBLEM-SOLVING AND DATA ANALYSIS

Mixed

03

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

Jeremy deposited x dollars in his investment account on January 1, 2001. The amount of money in the account doubled each year until Jeremy had 480 dollars in his investment account on January 1, 2005. What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

The number a is 70% less than the positive number b. The number c is 80% greater than a. The number c is how many times b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 ALGEBRA

Oil and gas production in a certain area dropped from 4 million barrels in 2000 to 1.9 million barrels in 2013. Assuming that the oil and gas production decreased at a constant rate, which of the following linear functions f best models the production, in millions of barrels, t years after the year 2000?

A. $f(t) = \frac{21}{130}t + 4$

B. $f(t) = \frac{19}{130}t + 4$

C. $f(t) = -\frac{21}{130}t + 4$

D. $f(t) = -\frac{19}{130}t + 4$

Q4 ALGEBRA

$$y < 6x + 2$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	20
5	32
7	44

B.

x	y
3	16
5	36
7	40

C.

x	y
3	16
5	28
7	40

D.

x	y
3	24
5	36
7	48

Q5 ALGEBRA

$$\frac{3}{5}x + \frac{3}{4}y = 7$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
1	$\frac{113}{20}$
2	$\frac{101}{20}$
4	$\frac{77}{20}$

B.

x	y
1	$\frac{47}{5}$
2	$\frac{44}{5}$
4	$\frac{38}{5}$

C.

x	y
1	$\frac{148}{15}$
2	$\frac{136}{15}$
4	$\frac{112}{15}$

D.

x	y
1	$\frac{128}{15}$
2	$\frac{116}{15}$

4	$\frac{92}{15}$

Q6 GRID-IN GEOMETRY AND TRIGONOMETRY

In triangle JKL, $\cos K = \frac{24}{51}$ and angle J is a right angle. What is the value of $\cos L$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GEOMETRY AND TRIGONOMETRY

Rectangles ABCD and EFGH are similar. The length of each side of EFGH is 6 times the length of the corresponding side of ABCD. The area of ABCD is 54 square units. What is the area, in square units, of EFGH?

A. 9

B. 36

C. 324

D. 1,944

Q8

ADVANCED MATH

The equation $\frac{x^2+6x-7}{x+7}=ax+d$ is true for all $x\neq -7$, where a and d are integers. What is the value of $a+d$?

A. -6

B. -1

C. 0

D. 1

Q9

GRID-IN ADVANCED MATH

$$y = x^2 - 14x + 22$$

The given equation relates the variables x and y. For what value of x does the value of y reach its minimum?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 ADVANCED MATH

$$fx = ax^2 + 4x + c$$

In the given quadratic function, a and c are constants. The graph of $y = fx$ in the xy-plane is a parabola that opens upward and has a vertex at the point h, k, where h and k are constants. If $k < 0$ and $f\text{-}9 = f3$, which of the following must be true?

I. $c < 0$
II. $a \geq 1$

A. I only

B. II only

C. I and II

D. Neither I nor II